完3.分子结构Molecular Structure

- 化学键Chemical Bonds
  - 离子键Ionic
    - 离子间的静电吸引Electrostatic attraction between ions
  - 共价键Covalent
    - 共用电子Sharing of electrons
  - 金属键Metallic
    - 金属原子与其他几个原子结合Metal atoms bonded to several other atoms
- 路易斯符号Lewis Symbols
  -

    Table 8.1 Lewis Symbols

Group	Element	Electron Configuration	Lewis Symbol	Element	Electron Configuration	Lewis Symbol
1A	Li	[He]$2s^1$	Li·	Na	[Ne]$3s^1$	Na·
2A	Be	[He]$2s^2$	·Be·	Mg	[Ne]$3s^2$	·Mg·
3A	B	[He]$2s^22p^1$	·Ḃ·	Al	[Ne]$3s^23p^1$	·Ȧl·
4A	C	[He]$2s^22p^2$	·Ċ·	Si	[Ne]$3s^23p^2$	·Ṡi·
5A	N	[He]$2s^22p^3$	·Ṅ:	P	[Ne]$3s^23p^3$	·Ṗ:
6A	O	[He]$2s^22p^4$	:Ȯ:	S	[Ne]$3s^23p^4$	:Ṡ:
7A	F	[He]$2s^22p^5$	·F̈:	Cl	[Ne]$3s^23p^5$	·C̈l:
8A	Ne	[He]$2s^22p^6$	:N̈e:	Ar	[Ne]$3s^23p^6$	:Ӓr:

  - 路易斯符号用一个点表示元素符号周围的每个价电子Lewis symbols use one dot for every valence electron around the element symbol.
  - 当形成化合物时，原子倾向于获得、失去或共享电子，直到它们被8个价电子包围(八隅体规则)When forming compounds, atoms tend to gain, lose, or share electrons until they are surrounded by eight valence electrons (the octet rule)
- 离子的形成Ionic Formation
  - 原子倾向于失去(金属)或获得(非金属)电子，使它们与稀有气体具有等电子Atoms tend to lose (metals) or gain (nonmetals) electrons to make them isoelectronic to the noble gases.
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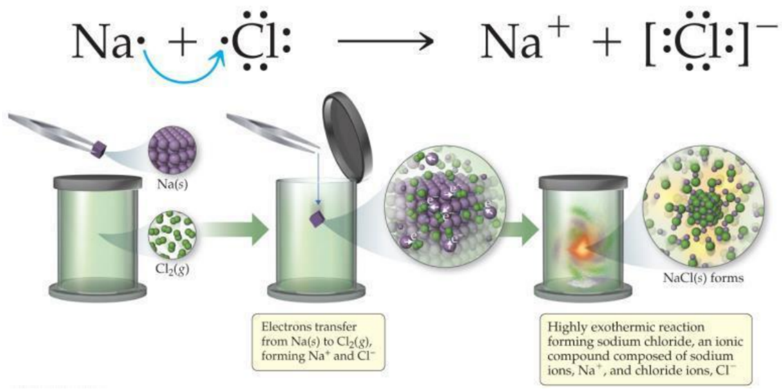

- 离子键的能量，玻恩-哈伯循环Energetics of Ionic Bonding—Born-Haber Cycle
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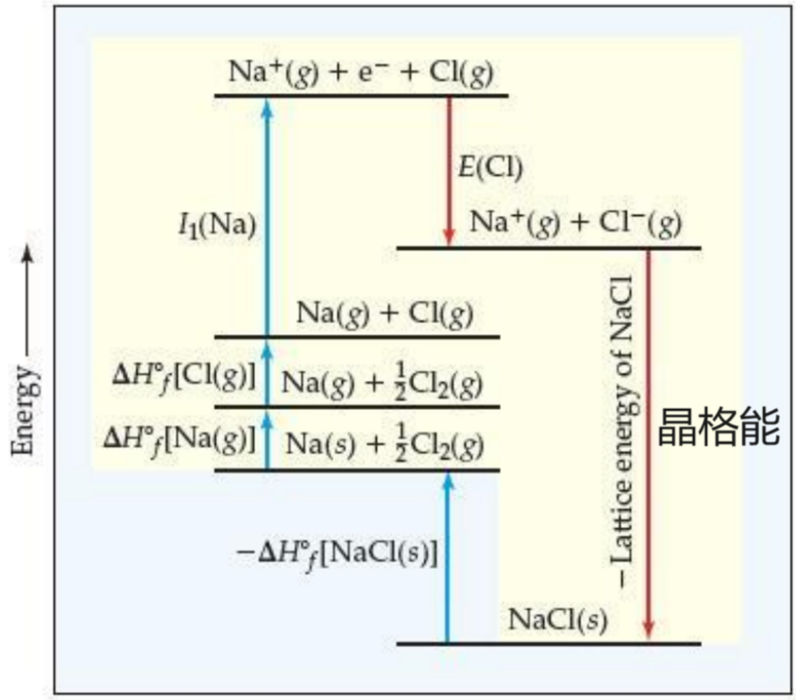

  - 从金属和非金属元素开始:Na(s)和Cl(g) Start with the metal and nonmetal elements: Na(s) and Cl(g).
  - 生成气态原子:Na(g)和Cl(g) Make gaseous atoms: Na(g) and Cl(g).
  - 生成离子:Na+(g)和Cl-(g) Make ions: Na+(g) and Cl-(g).
  - 结合离子:NaCl(s) Combine the ions: NaCl(s).
  - 我们已经讨论过制造离子(电离能和电子亲和能)We already discussed making ions (ionization energy and electron affinity).
  - 元素转化为原子需要能量(吸热)It takes energy to convert the elements to atoms. (**endothermic**)
  - 生成阳离子需要能量(吸热)It takes energy to create a cation (**endothermic**).
  - 能量通过制造阴离子而释放(放热)Energy is released by making the anion(**exothermic**)
  - 固体的形成释放出大量的能量(放热)The formation of the solid releases a huge amount of energy (**exothermic**).
  - 这使得元素放热生成盐This makes the formation of salts from the elements exothermic.
- 晶格能Lattice Energy

- 将一摩尔固体离子化合物完全分离成气态离子所需的能量the energy required to completely separate a mole of a solid ionic compound into its gaseous ions.
- 晶格能越大，离子化合物越稳定The greater the lattice energy, the more stable the ionic compound.
- 与静电相互作用有关的能量受库仑定律支配The energy associated with electrostatic interactions is governed by Coulomb's law

> $$E_{el} = \kappa \frac{Q_1 Q_2}{d}$$
>
> ***Q1 and Q2***: the charges on the ions (measured in Coulombs)
> ***d*** : The distance between their centers
> ***k***: a constant (8.99×10^9 J-m/C^2)

- 晶格能随着~而增大Lattice energy increases with
  - 离子电荷增加increasing charge on the ions
  - 离子减小decreasing size of ions
  -

    Table 8.2 Lattice Energies for Some Ionic Compounds

Compound	Lattice Energy (kJ/mol)	Compound	Lattice Energy (kJ/mol)
LiF	1030	$MgCl_2$	2326
LiCl	834	$SrCl_2$	2127
LiI	730		
NaF	910	MgO	3795
NaCl	788	CaO	3414
NaBr	732	SrO	3217
NaI	682		
KF	808	ScN	7547
KCl	701		
KBr	671		
CsCl	657		
CsI	600		

- 共价键Covalent Bonding
  - 在共价键中，原子共用电子In covalent bonds, atoms share electrons.
  - 在这些键中有几种静电相互作用There are several electrostatic interactions in these bonds:
    - 电子和原子核之间的吸引力attractions between electrons and nuclei
    - 电子之间的排斥力repulsions between electrons
    - 原子核间的斥力repulsions between nuclei
  - 要形成一个键，引力必须大于斥力For a bond to form, the attractions must be greater than the repulsions.
- 路易斯结构Lewis Structures
  - 共用电子形成共价键可以用路易斯结构来证明Sharing electrons to make covalent bonds can be demonstrated using Lewis structures.

- 我们首先试着通过共享电子给每个原子8个电子(例外:氢)We start by trying to give each atom 8 electrons by sharing electrons.(exception: Hydrogen)
- 最简单的例子是氢H2和氯Cl2，如下所示The simplest examples are for hydrogen, H2, and chlorine, Cl2, shown below.

  

- 路易斯结构上的电子Electrons on Lewis Structures
  - 孤对电子Lone pairs
    - 在路易斯结构中只位于一个原子上的电子electrons located on only one atom in a Lewis structure
  - 成键电子对Bonding pairs
    - 共享电子;它们可以用两个点或一条线来表示shared electrons; they can be represented by two dots or one line
- 重键Multiple Bonds
  - 一对电子=单键one pair of electrons = single bonds.
  - 两对电子=双键two pairs of electrons = double bonds.
  - 有时，两个原子共用三个化学键，它们被称为三键Sometimes, three bonds are shared between two atoms. These are called triple bonds.
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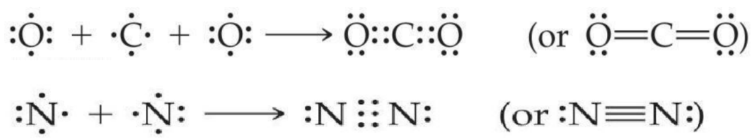
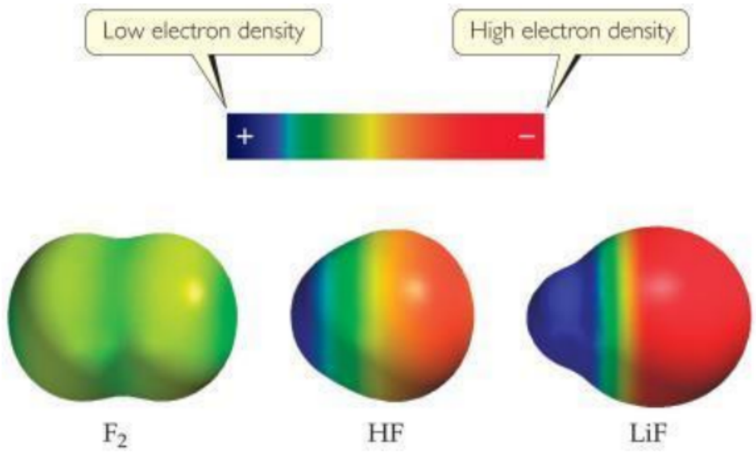
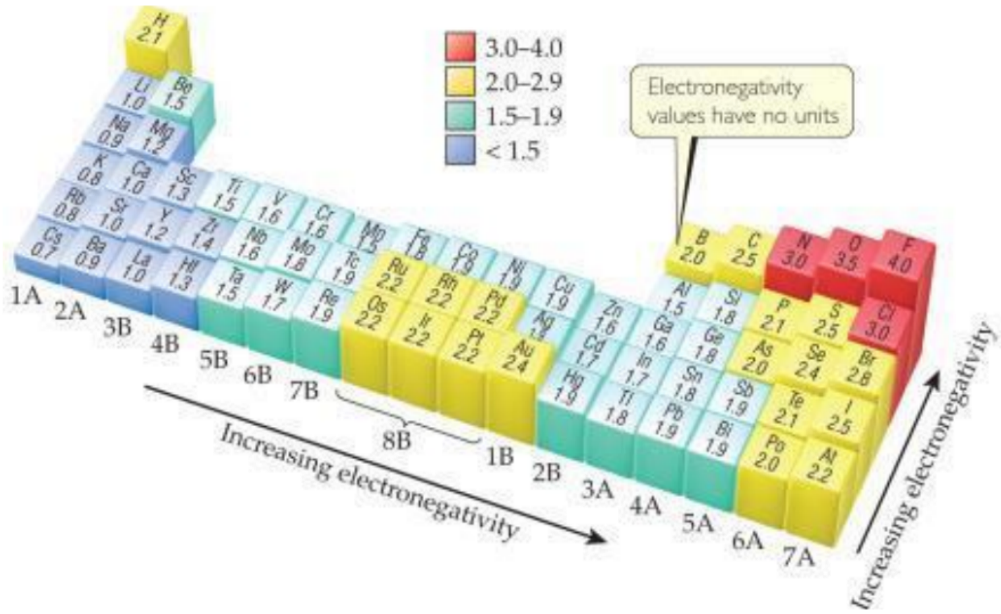
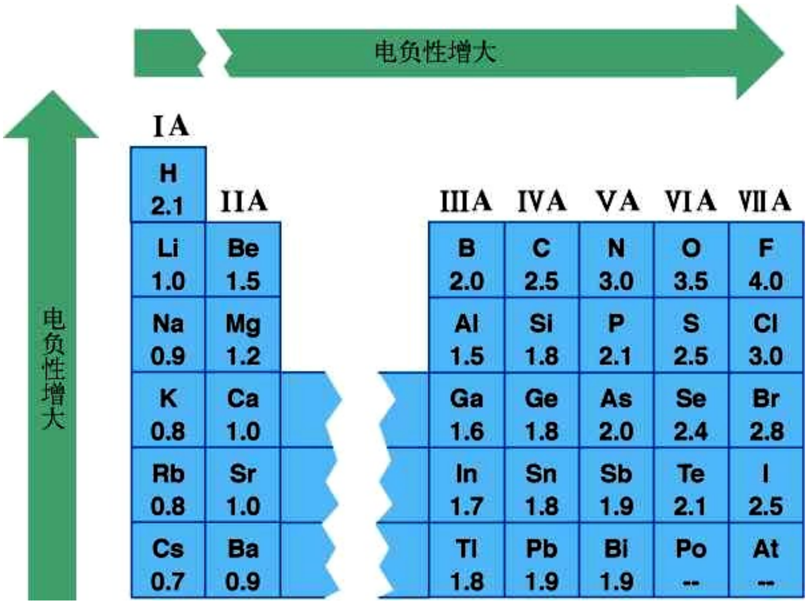
    

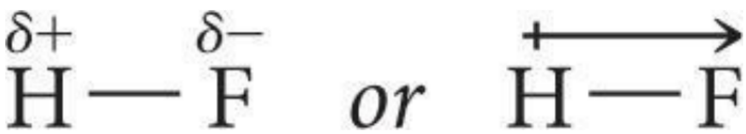
- 极性共价键Polar Covalent Bonds
  - 共价键中的电子并不总是平均共用的The electrons in a covalent bond are not always shared equally.
  - 氟对它与氢共享的电子的吸引力比氢更大Fluorine pulls harder on the electrons it shares with hydrogen than hydrogen does.
  - 因此，分子的氟端比氢端有更大的电子密度Therefore, the fluorine end of the molecule has more electron density than the hydrogen end.
  -



- 电负性Electronegativity
  - 电负性是指分子中原子吸引电子的能力Electronegativity is the ability of an atom in a molecule to attract electrons to itself.
  - 电负性随~而增长electronegativity generally increases...
    - 一个周期从左到右from left to right across a period.
    - 一个族从下到上from the bottom to the top of a group.
  - 图



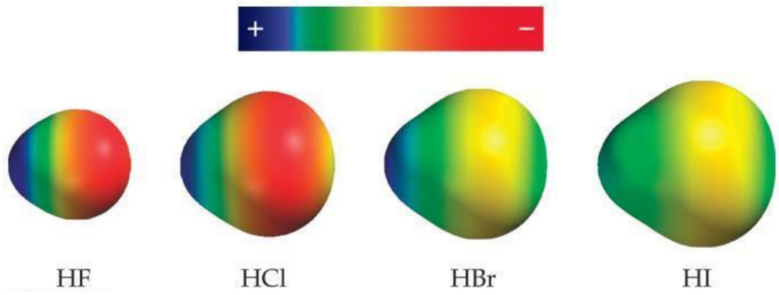


- 电负性和极性共价键Electronegativity and Polar Covalent Bonds
  - 当两个原子不平等地共享电子时，就形成了一个极性共价键When two atoms share electrons unequally, a polar covalent bond results.
  - 电子倾向于在电负性更强的原子周围停留更长的时间，结果是部分负电荷偏移(而不是完全的电荷转移)，用δ-表示Electrons tend to spend more time around the more electronegative atom. The result is a partial negative charge (not a complete transfer of charge). It is represented byδ-
  - 另一个原子的正电荷更大，用δ+表示The other atom is "more positive," orδ+
  - 

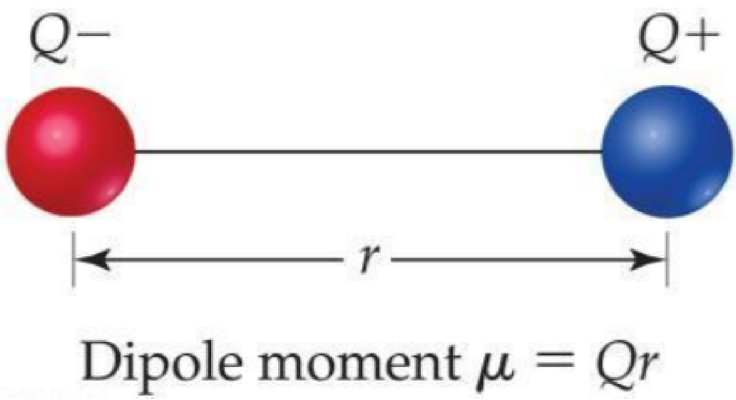
- 极性共价键Polar Covalent Bonds
  - 电负性差异越大，键的极性越强The greater the difference in electronegativity, the more polar is the bond.
  - 图

Table 8.3 Bond Lengths, Electronegativity Differences, and Dipole Moments of the Hydrogen Halides

Compound	Bond Length (Å)	Electronegativity Difference	Dipole Moment (D)
HF	0.92	1.9	1.82
HCl	1.27	0.9	1.08
HBr	1.41	0.7	0.82
HI	1.61	0.4	0.44



- 偶极子Dipoles
  - 当两个相等但方向相反的电荷相隔一段距离时，偶极子就形成了When two equal, but opposite, charges are separated by a distance, a dipole forms.
  - 偶极矩，μ，由两个相等但相反的电荷产生，间隔距离r，计算如下A **dipole moment**, μ, produced by two equal but opposite charges separated by a distance, r, is calculated
    - 偶极矩是由磁偶极或电偶极产生的力矩，特别是指两极之间距离和任一极的大小的乘积
    - μ= Qr
    - 单位是德拜It is measured in debyes (D)
    -



- 判断离子化合物与共价化合物Is a Compound Ionic or Covalent?
  - 最简单的方法:金属+非金属是离子;非金属+非金属是共价的Simplest approach: Metal+ nonmetal is ionic; nonmetal+ nonmetal is covalent.
  - 也有很多例外:它不考虑金属的氧化值(更高的氧化值可以形成共价键)There are many exceptions: It doesn't take into account oxidation number of a metal (higher oxidation numbers can give covalent bonding).
  - 电负性差可用;这个表仍然没有考虑氧化值Electronegativity difference can be used; the table still doesn't take into account oxidation number.
  - 化合物的性质通常是最好的:例如，熔点较低意味着共价键Properties of compounds are often best: Lower melting points mean covalent bonding, for example.
  - 图

Classification of bonds by difference in electronegativity
根据电负性的差异对键进行分类

Difference	Bond Type
0	Nonpolar Covalent 非极性共价键
≥ 2	Ionic 离子键
0 < and <2	Polar Covalent 极性共价键

电负性差逐渐增大
Increasing difference in electronegativity →

Nonpolar Covalent → Polar Covalent → Ionic

share e^- | partial transfer of e^- | transfer e^-

- 书写路易斯结构Writing Lewis Structures (Covalent Molecules)
  - 图

Writing Lewis Structures (Covalent Molecules)

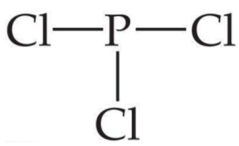
PCl_3

1. Sum the valence electrons from all atoms, taking into account overall charge.
   - If it is an anion, add one electron for each negative charge.
   - If it is a cation, subtract one electron for each positive charge.

Keep track of the electrons:

5 + 3(7) = 26

Writing Lewis Structures

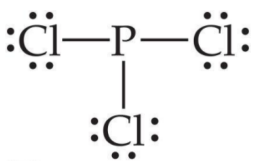


2. Write the symbols for the atoms, show which atoms are attached to which, and connect them with a single bond (a line representing two electrons).

Keep track of the electrons:

26 − 6 = 20

Writing Lewis Structures

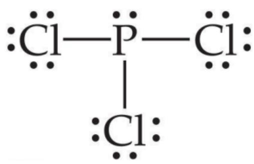


3. Complete the octets around all atoms bonded to the central atom.

Keep track of the electrons:

26 − 6 = 20; 20 − 18 = 2

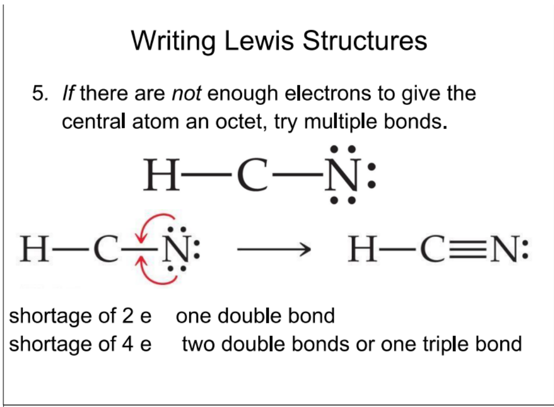
Writing Lewis Structures

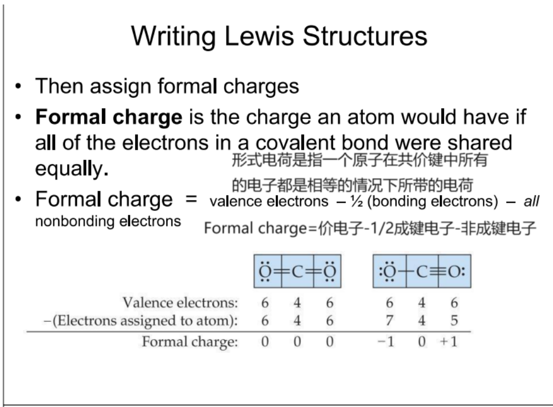


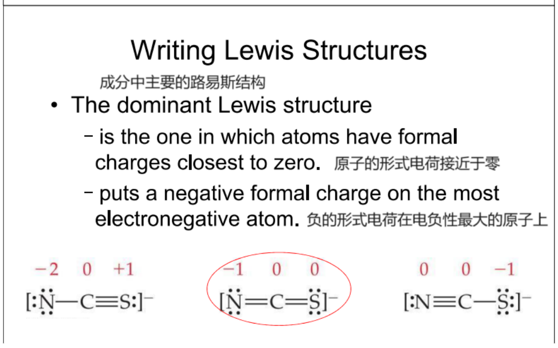
4. Place any leftover electrons on the central atom.

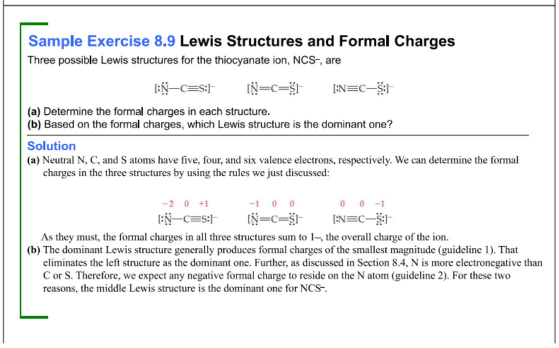
Keep track of the electrons:

26 − 6 = 20; 20 − 18 = 2; 2 − 2 = 0



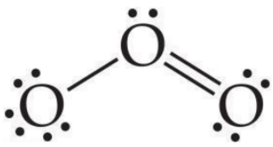




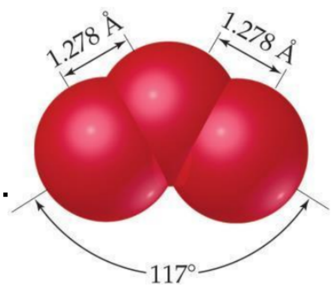


- 共振结构
  - 臭氧的结构
    - 根据规则应该是上图，但自然中氧之间的键是相同的

      Following our rules, this is the Lewis structure we would draw for ozone, O_3.

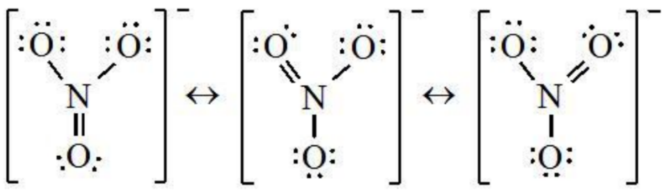
      

      However, it doesn't agree with what is observed in nature: Both O to O connections are the same.

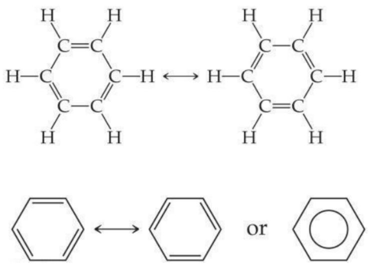
      

  - 一个路易斯结构不能准确地描述像臭氧这样的分子One Lewis structure cannot accurately depict a molecule like ozone.
  - 我们用多种结构，共振结构，来描述分子We use multiple structures, resonance structures, to describe the molecule.
  - 当一个分子、离子或自由基的结构不能用路易斯结构式正确地描述时，可以用多个路易斯式表示，这些路易斯式称为共振结构(resonance structure，又称极限式或正则结构)，在共振结构之间用双箭头“↔”联系，以表示它们的共振关系

$NO^{3-} = 24e^-$

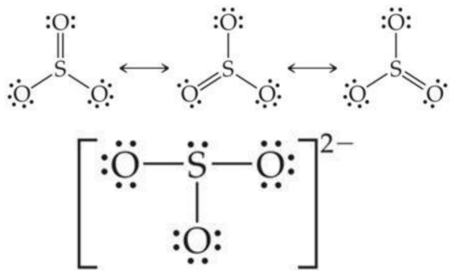


- 苯
    - 图

    - 有机化合物苯C6H6有两种共振结构The organic compound benzene,C6H6， has two resonance structures.
    - 它通常被描绘成一个六边形，里面有一个圆圈来表示环中的离域电子It is commonly depicted as a hexagon with a circle inside to signify the delocalized electrons in the ring.
    - 定域电子专门在一个原子上或在两个原子之间共享;离域电子由多个原子共享Localized electrons are specifically on one atom or shared between two atoms; Delocalized electrons are shared by multiple atoms.

- 哪个硫氧键更短，SO3还是SO3^2-? Which is predicted to have the shorter sulfur—oxygen bonds, SO3 or SO3^2-?
    - 图

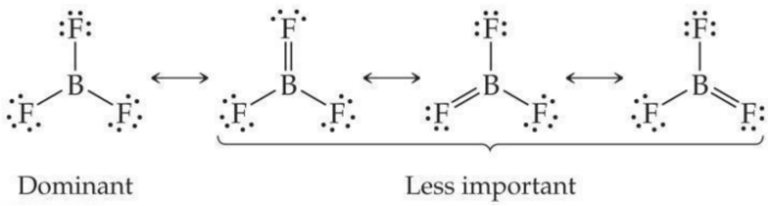
        

    - SO3的实际结构是这三种物质的均匀混合，也就是说，S-O应该比单键短，但没有双键短the actual structure of SO3 is an equal blend of all three. That is, S—O should be shorter than single bonds but not as short as double bonds.
    - SO3^2-离子有26个电子，这导致了主导的路易斯结构，其中所有的S-0键都是单键The SO3^2- ion has 26 electrons, which leads to a dominant Lewis structure in which all the S—O bonds are single
    - SO3应该有较短的S-O键SO3^2应该有较长的键SO3 should have the shorter S—O bonds and SO3^2- the longer ones.

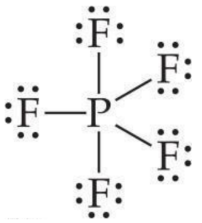
- 8电子规则(八隅体规则)的例外Exceptions to the **Octet Rule**
  - [奇电子分子]带有奇数电子的离子或分子ions or molecules with an odd number of electrons
    - 虽然相对罕见，而且通常相当不稳定和活泼，但还是有一些离子和分子带有奇数个电子Though relatively rare and usually quite unstable and reactive, there are ions and molecules with an odd number of electrons.

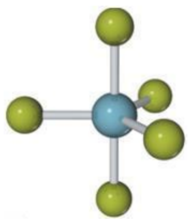
      

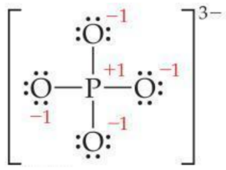
  - [缺电子分子]小于一个八隅体的离子或分子ions or molecules with less than an octet
    - 在碳之前的第二周期元素可以生成少于8个电子的稳定化合物Elements in the second period before carbon can make stable compounds with fewer than eight electrons.
    - 考虑BF3 Consider BF3
      - 给硼一个填满的八隅体，硼带一个负电荷，氟带一个正电荷Giving boron a filled octet places a negative charge on the boron and a positive charge on fluorine.
      - 这并不是BF3中电子分布的准确图像This would not be an accurate picture of the distribution of electrons in BF3.
      - 图

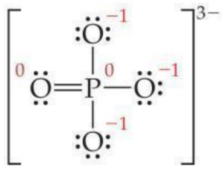
        

      - 如果填充中心原子的八隅体导致中心原子带负电荷而电负性更强的外层原子带正电荷，就不要填充中心原子的八隅体If filling the octet of the central atom results in a negative charge on the central atom and a positive charge on the more electronegative outer atom, don't fill the octet of the central atom.
  - [超价分子]价电子多于8个的离子或分子(扩大的八隅体)ions or molecules with more than eight valence electrons (an expanded octet)
    - 当元素在周期表中处于第3或往下的周期时(例如，第3、4、5周期等)，它可以使用d轨道形成4个以上的键When an element is in period 3 or below in the periodic table (e.g., periods 3, 4, 5, etc.), it can use d-orbitals to make more than four bonds.
    - 例如:PF5和磷酸盐Examples: PF5 and phosphate
      - 注意:磷酸实际上有四个共振结构，在P原子上有五个键Note: Phosphate will actually have four resonance structures with five bonds on the P atom
      - 图









- 共价键强度Covalent Bond Strength
  - 最简单地说，键的强度是通过确定断开键所需的能量来衡量的Most simply, the strength of a bond is measured by determining how much energy is required to break the bond.
  - 这叫做键焓This is called the **bond enthalpy**.
  - The bond enthalpy for a Cl—Cl bond, D(Cl—Cl), is measured to be 242 kJ/mol.
  - We write out reactions for breaking one mole of those bonds: Cl—Cl→2Cl·
  - 平均键焓Average Bond Enthalpies
    - 平均键焓是正的，因为断键是一个吸热过程Average bond enthalpies are positive，because bond breaking is an **endothermic** process
    - 注意，这些是许多不同化合物的平均值;在自然界中，并不是一对原子的每个键都有完全相同的键能Note that these are averages over many different compounds; not every bond in nature for a pair of atoms has exactly the same bond energy.
    - 图

      Table 8.4 Average Bond Enthalpies (kJ/mol)

Single Bonds							
C—H	413	N—H	391	O—H	463	F—F	155
C—C	348	N—N	163	O—O	146		
C—N	293	N—O	201	O—F	190	Cl—F	253
C—O	358	N—F	272	O—Cl	203	Cl—Cl	242
C—F	485	N—Cl	200	O—I	234		
C—Cl	328	N—Br	243			Br—F	237
C—Br	276			S—H	339	Br—Cl	218
C—I	240	H—H	436	S—F	327	Br—Br	193
C—S	259	H—F	567	S—Cl	253		
		H—Cl	431	S—Br	218	I—Cl	208
Si—H	323	H—Br	366	S—S	266	I—Br	175
Si—Si	226	H—I	299			I—I	151
Si—C	301						
Si—O	368						
Si—Cl	464						
Multiple Bonds							
C=C	614	N=N	418	O=O	495		
C≡C	839	N≡N	941				
C=N	615	N=O	607	S=O	523		
C≡N	891			S=S	418		
C=O	799						
C≡O	1072						

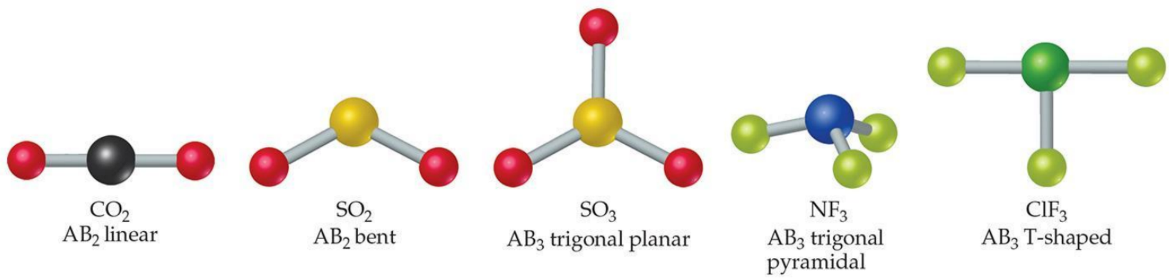
    - 一个反应的焓变=化学键断裂的焓变-化学键生成的焓变
    - ΔHrxn = Σ(bond enthalpies of all bonds broken)-Σ(bond enthalpies of all bonds formed)
    - 我们还可以测量不同键类型的平均键长We can also measure an average bond length for different bond types.

- 随着两个原子间化学键数的增加，化学键的长度减小As the number of bonds between two atoms increases, the bond length decreases.
- 图

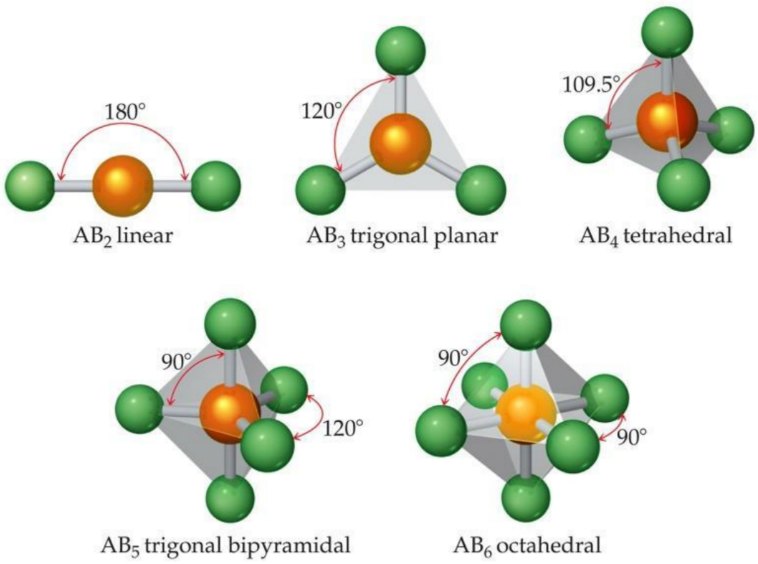
Table 8.5 Average Bond Lengths for Some Single, Double, and Triple Bonds

Bond	Bond Length (Å)	Bond	Bond Length (Å)
C—C	1.54	N—N	1.47
C=C	1.34	N=N	1.24
C≡C	1.20	N≡N	1.10
C—N	1.43	N—O	1.36
C=N	1.38	N=O	1.22
C≡N	1.16		
		O—O	1.48
C—O	1.43	O=O	1.21
C=O	1.23		
C≡O	1.13		

- 分子形状Molecular Shapes
  - 路易斯结构显示成键和孤对，但不表示形状Lewis structures show bonding and lone pairs but do not denote shape.
  - 然而，我们使用路易斯结构来帮助我们确定形状However, we use Lewis structures to help us determine shapes.
  - 这里我们看到了两个或三个原子与一个中心原子相连的分子的一些常见形状Here we see some common shapes for molecules with two or three atoms connected to a central atom.



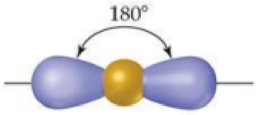
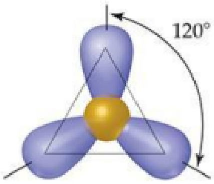
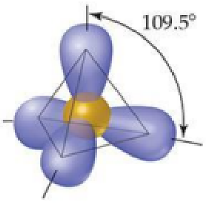
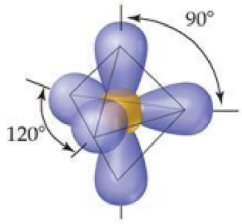
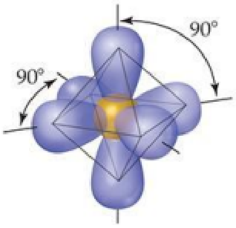
- VSEPR
  - 键角和键长决定了分子的形状和大小The bond angles and bond lengths determine the shape and size of molecules.
  - 电子对相互排斥Electron pairs repel each other.
  - Electron pairs are as far apart as possible; this allows predicting the shape of the molecule.电子对之间的距离越远越好;这就可以预测分子的形状
  - This is the valence-shell electron-pair repulsion(VSEPR) model.这是价电子层电子对斥力(VSEPR)模型
  - 图



- Electron Domains电子域
  - We can refer to the directions to which electrons point as electron domains. This is true whether there is one or more electron pairs pointing in that direction. It is also true if it is a lone pair or a bond.我们可以把电子指向的方向称为电子域，无论有一个或多个电子对指向那个方向，这都是正确的，如果是孤对或键，也成立
  - The central atom in this molecule, A, has four electron domains.分子的中心原子A，有四个电子域

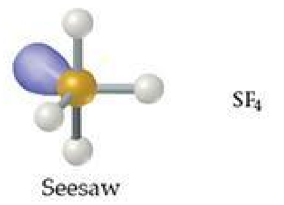
- 电子域几何electron-domain geometry
  - The table shows the electron-domain geometries for two through six electron domains around a central atom.该表显示了围绕中心原子的2到6个电子域的电子域几何形状

TABLE 9.1 Electron-Domain Geometries as a Function of Number of Electron Domains

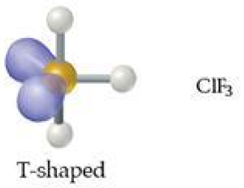
Number of Electron Domains*	Arrangement of Electron Domains	Electron Domain Geometry	Predicted Bond Angles
2		Linear	180°
3		Trigonal planar	120°
4		Tetrahedral	109.5°
5		Trigonal bipyramidal	120° 90°
6		Octahedral	90°

*The number of electron domains is sometimes called the *coordination number* of the atom.

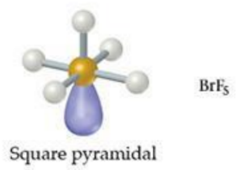
- To determine the electron-domain geometry, count the total number of lone pairs, single, double, and triple bonds on the central atom.为了确定电子域的几何形状，计算中心原子上的孤对、单键、双键和三键的总数
- 好好看看英文怎么写
  - Linear线性
  - Trigonal planar平面三角形
  - Tetrahedral四面体
  - Trigonal bipyramidal三角双锥
  - Trigonal pyramidal三角锥
  - Octahedral八面体
  - Bent角形
  - Seesaw-shaped跷跷板型



- T-shaped



- Square pyramidal方形锥体，金字塔



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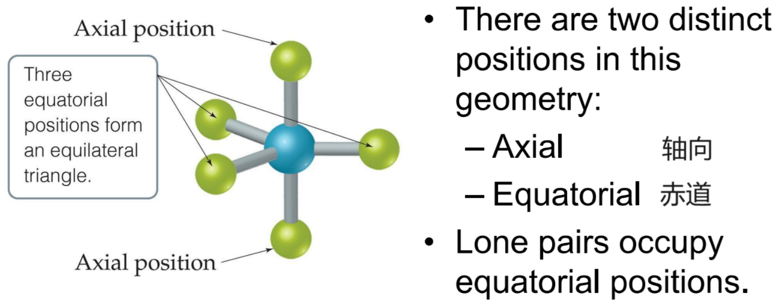
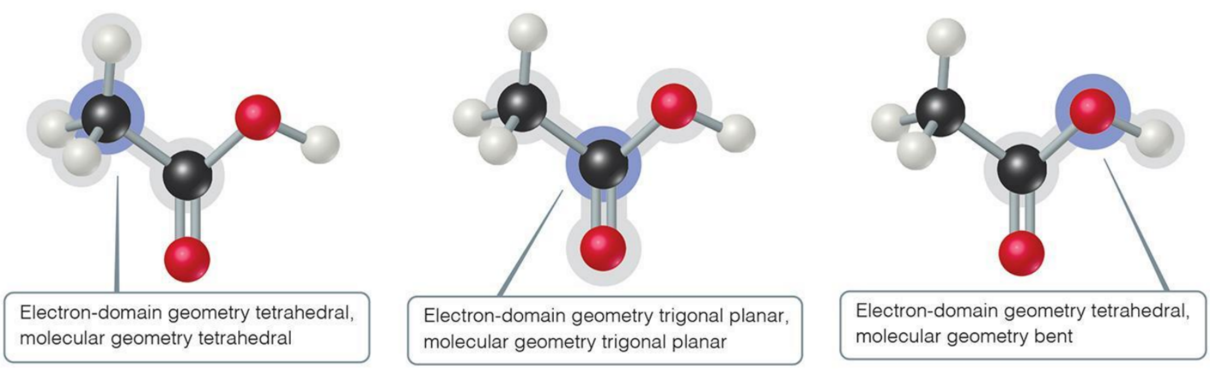
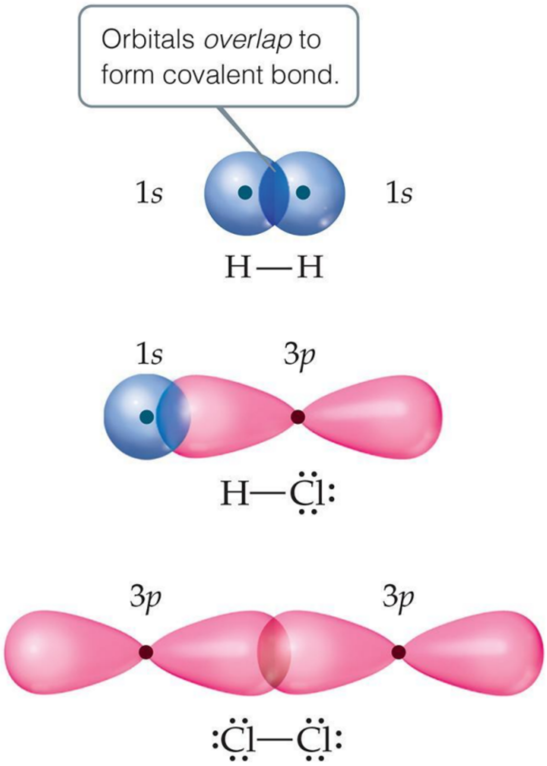


- Shapes of Larger Molecules大分子的形状
  - For larger molecules, look at the geometry about each atom rather than the molecule as a whole.对于较大的分子，要看每个原子的几何形状，而不是整个分子

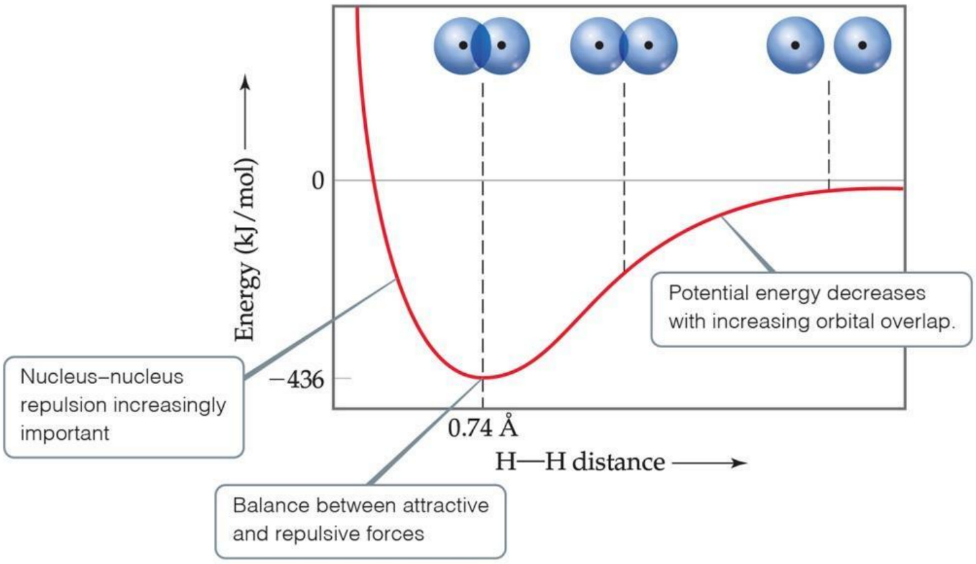


- 极性分子
  - 可以通过计算dipole moment偶极矩判断是否=0来判断是否是非极性分子
  - All diatomic molecules with polar bonds are polar所有带极性键的双原子分子都是极性的
- Valence-Bond Theory价键理论

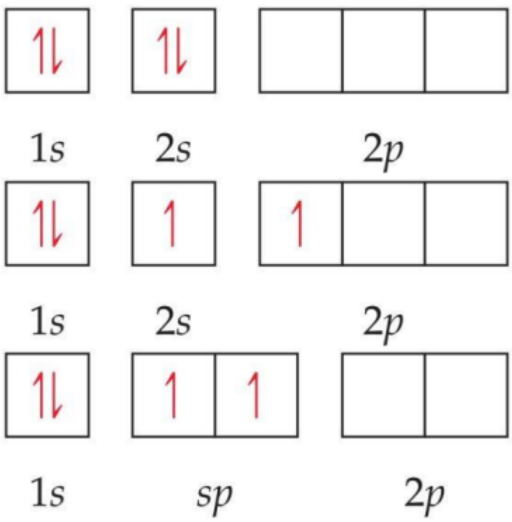
- In valence-bond theory, electrons of two atoms begin to occupy the same space.在价键理论中，两个原子的电子开始占据相同的空间
- This is called "overlap" of orbitals这就是所谓的“重叠”轨道
- The sharing of space between two electrons of opposite spin results in a covalent bond.两个自旋相反的电子共享空间形成共价键
- 图

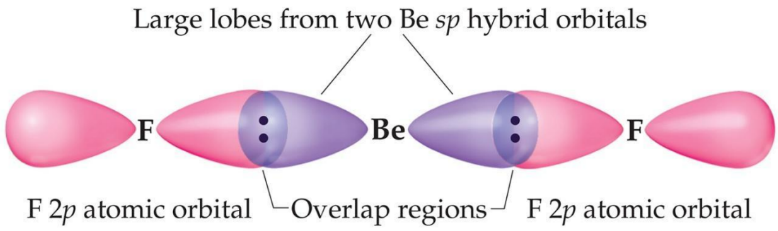
- 引斥力的平衡
  - Increased overlap brings the atoms together until a balance is reached between the like charge repulsions and the electron-nucleus attraction.增强的重叠使原子聚集在一起，直到在类似电荷的排斥和电子-原子核的吸引之间达到平衡
  - Atoms can't get too close because the internuclear repulsions get too great.原子不能靠得太近，因为原子核间的斥力太大了
  - 图



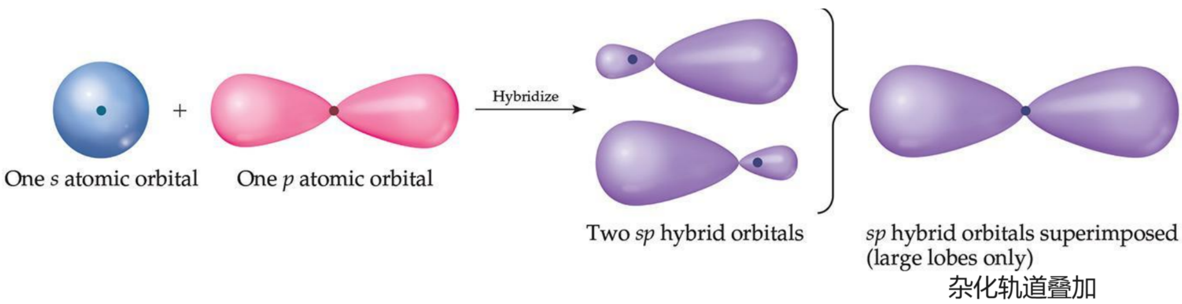
- Hybrid Orbitals杂化轨道
  - **Hybrid orbitals** form by "mixing" of atomic orbitals to create new orbitals of equal energy, called degenerate orbitals. 杂化轨道是通过原子轨道的“混合”形成能量相等的新轨道，称为简并轨道
  - This process is called **hybridization**.这个过程叫做杂化
  - When two orbitals "mix" they create twoorbitals; when three orbitals mix, they create three orbitals; and so on.当两个轨道“混合”时，就会产生两个轨道轨道;当三个轨道混合时，它们会形成三个轨道;等等......
  - Be的sp杂化轨道
    - When we look at the orbital diagram for beryllium (Be), we see that there are only paired electrons in full sublevels.当我们看铍(Be)的轨道图时，我们看到在完整的亚能级中只有成对电子
    - Be makes electron-deficient compounds with two bonds for Be. Why? sp hybridization (mixing of ones orbital and one p orbital).Be生成有两个键的缺电子化合物。为什么?Sp杂化(1轨道和1 p轨道的混合)
    - 图



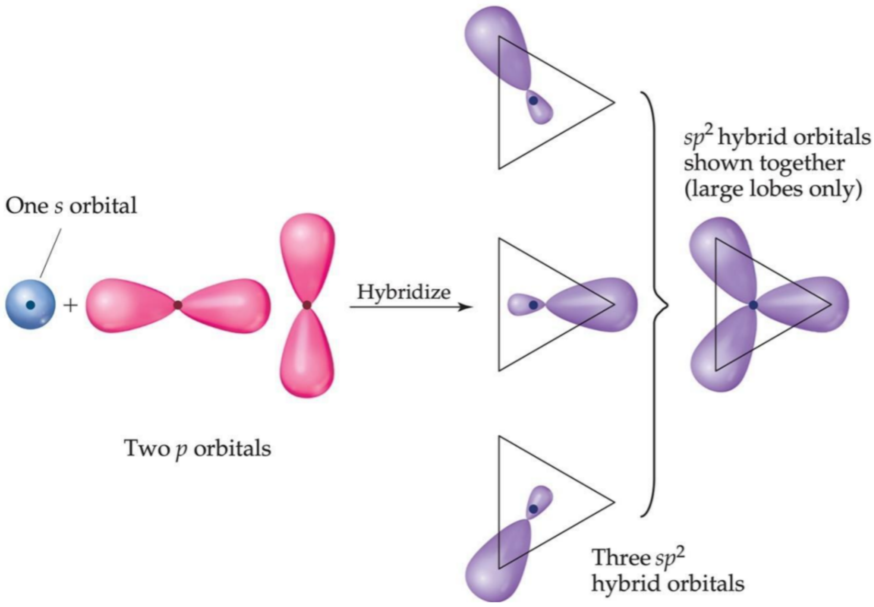
- These two degenerate orbitals would align themselves 180° from each other.这两个简并轨道会彼此排成180°的直线
- This is consistent with the observed geometry of Be compounds (like BeF2) and VSEPR: linear.这与观察到的Be化合物(如BeF2)和VSEPR的几何形状一致:线性
- 图



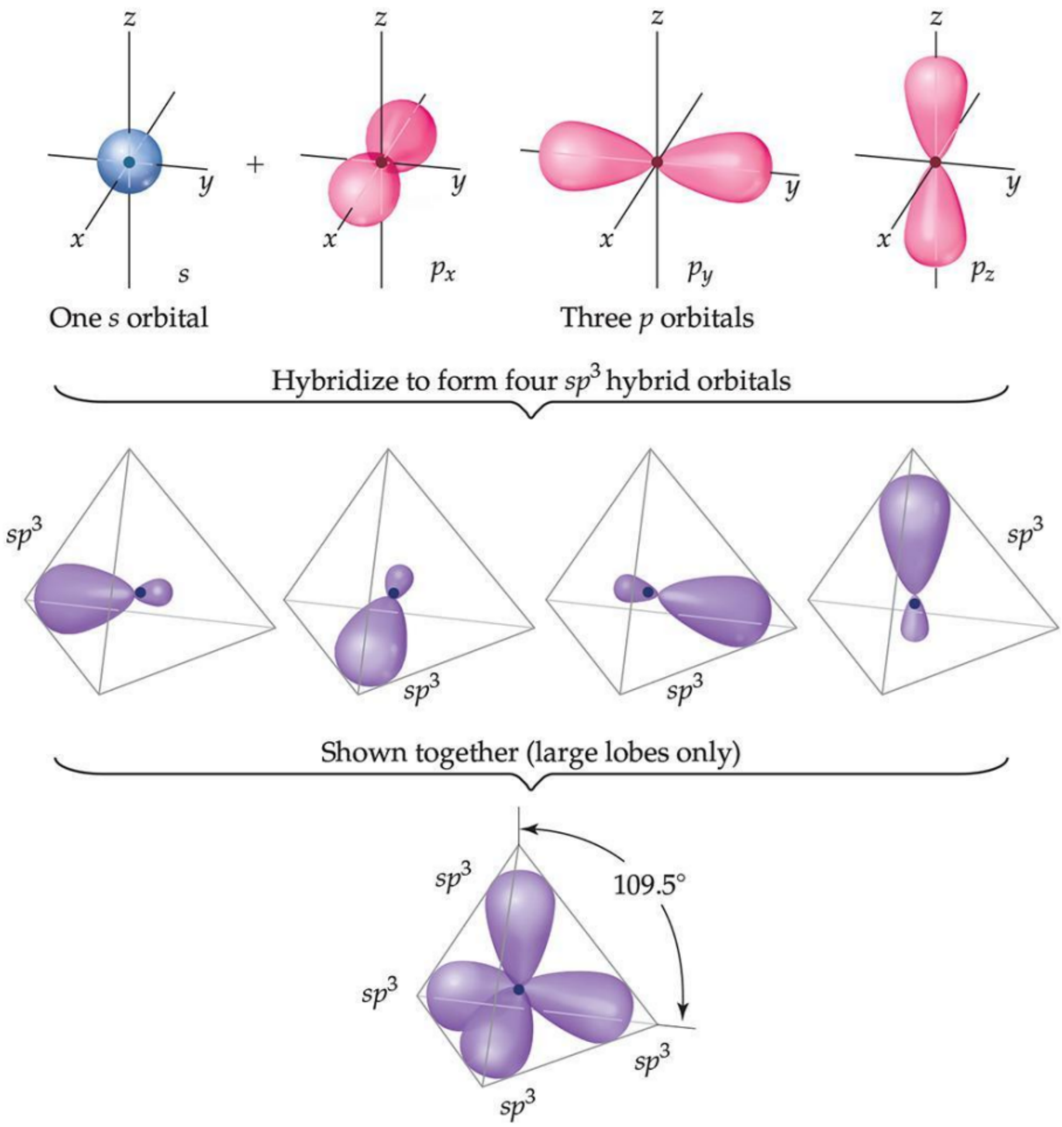
- sp轨道
  - Mixing the s and p orbitals yields two degenerate orbitals that are hybrids of the two orbitals.s轨道和p轨道混合会产生两个简并轨道，它们是这两个轨道的杂化轨道
  - The sp hybrid orbitals each have two lobes like a p orbital. sp杂化轨道和p轨道一样，都有两个叶
  - One of the lobes is larger and more rounded, as is the s orbital.其中一个叶瓣更大更圆，s轨道也是如此
  - 图



- B(Boron)的sp2杂化轨道
  - 图

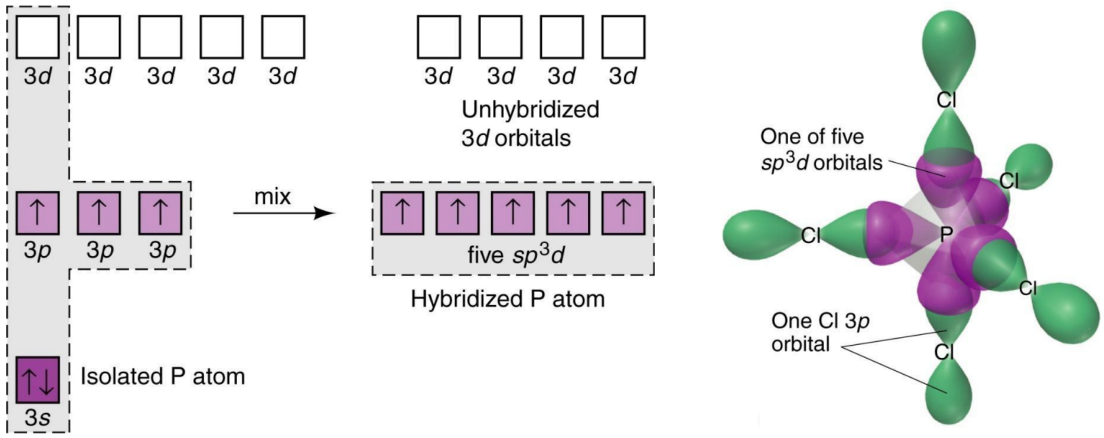


- C的sp3杂化轨道
  - 图

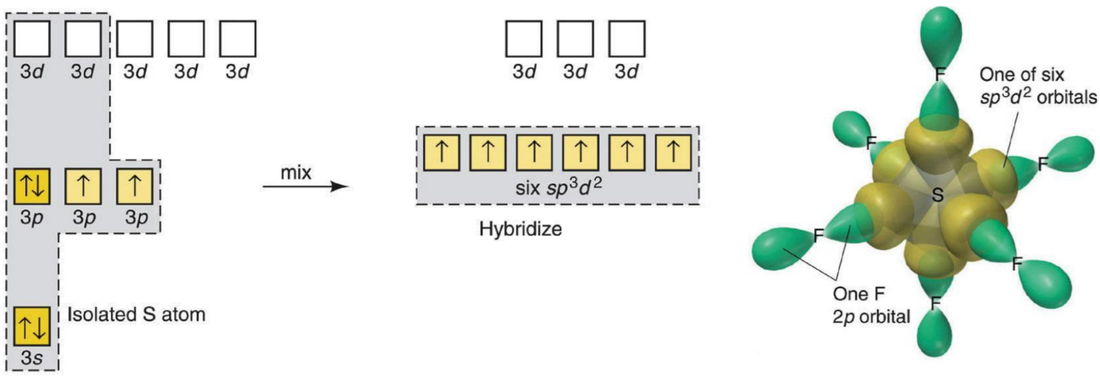


- PCl5中的sp3d杂化

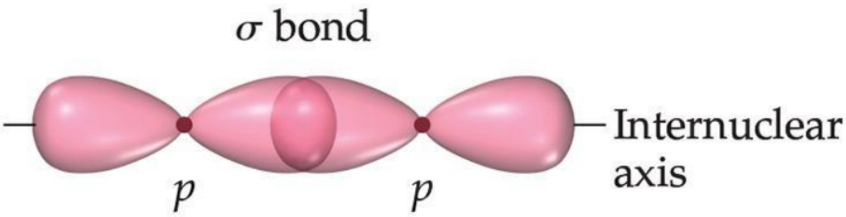
- The formation of more than four bonding orbitals requires d orbital involvement in hybridization.4个以上成键轨道的形成需要d轨道参与杂化
- 图



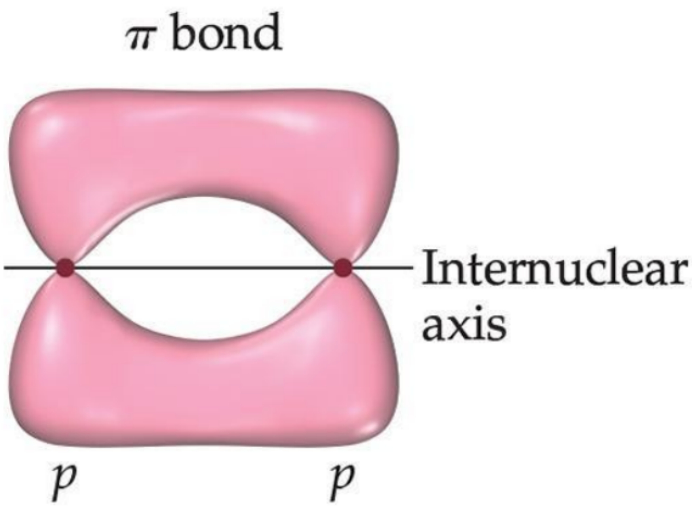
- SF6中的sp3d2杂化
  - 图



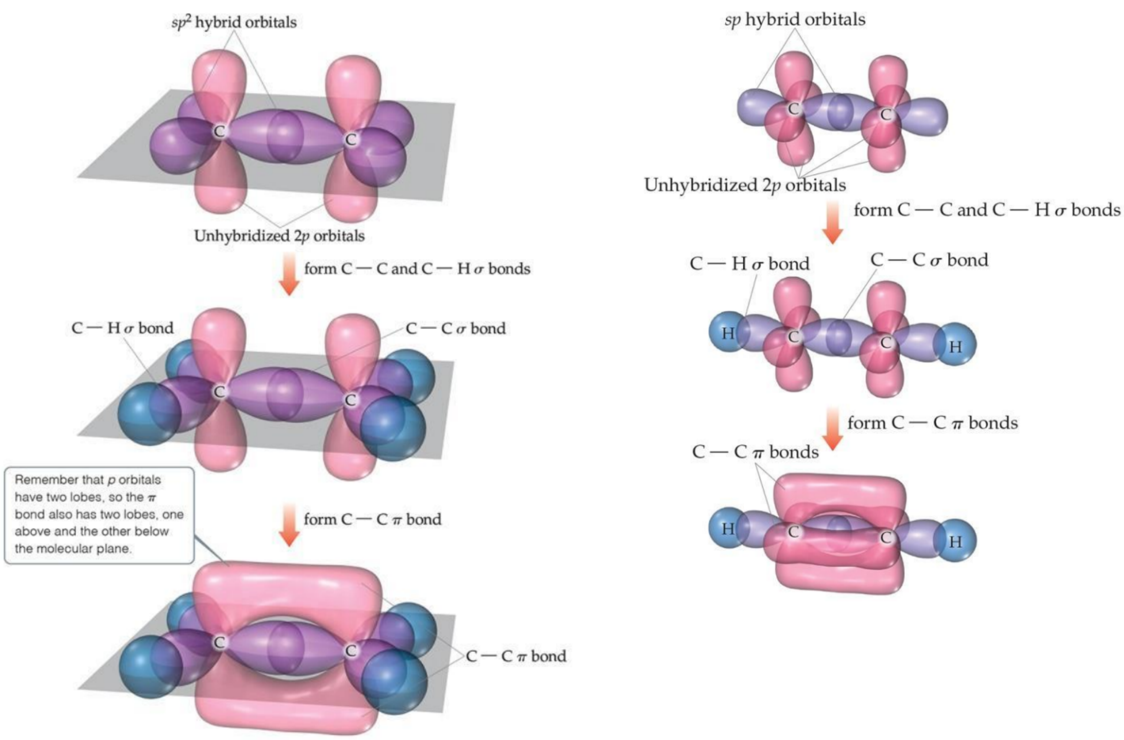
- Hypervalent Molecules超价分子
  - The elements that have more than an octet中心原子价层电子超过8
  - Valence-bond model would use d orbitals to make more than four bonds.价电子键模型会用d轨道形成4个以上的键
  - This view works for period 3 and below.此视角适用于周期3及往下(3，4，5)
  - Theoretical studies suggest that the energy needed would be too great for this.理论研究表明，这需要的能量太大了
- Types of Bonds共价键的种类
  - Sigma (σ) bond
    - head-to-head overlap头对头重叠
    - cylindrical symmetry of electron density about the internuclear axis.关于核间轴的电子密度的圆柱对称

- Pi (π) bond
    - sideways overlap.肩并肩重叠
    - mirror symmetry of electron density above and below the internuclear axis.核间轴上下的电子密度上下对称

- Single bonds are always σ-bonds单键都是σ键;Multiple bonds have one σ-bond; all other bonds are π-bonds.双键及以上有一个σ键;其它键都是π键

- 顺反异构

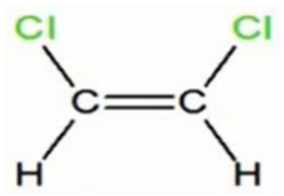
顺式与反式同分异构体

Cis Vs Trans Isomers

二氯乙烯

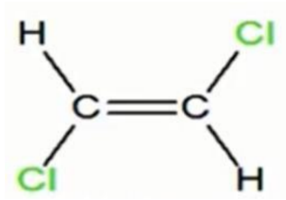
1,2- dichloroethene（$C_2H_2Cl_2$）

顺



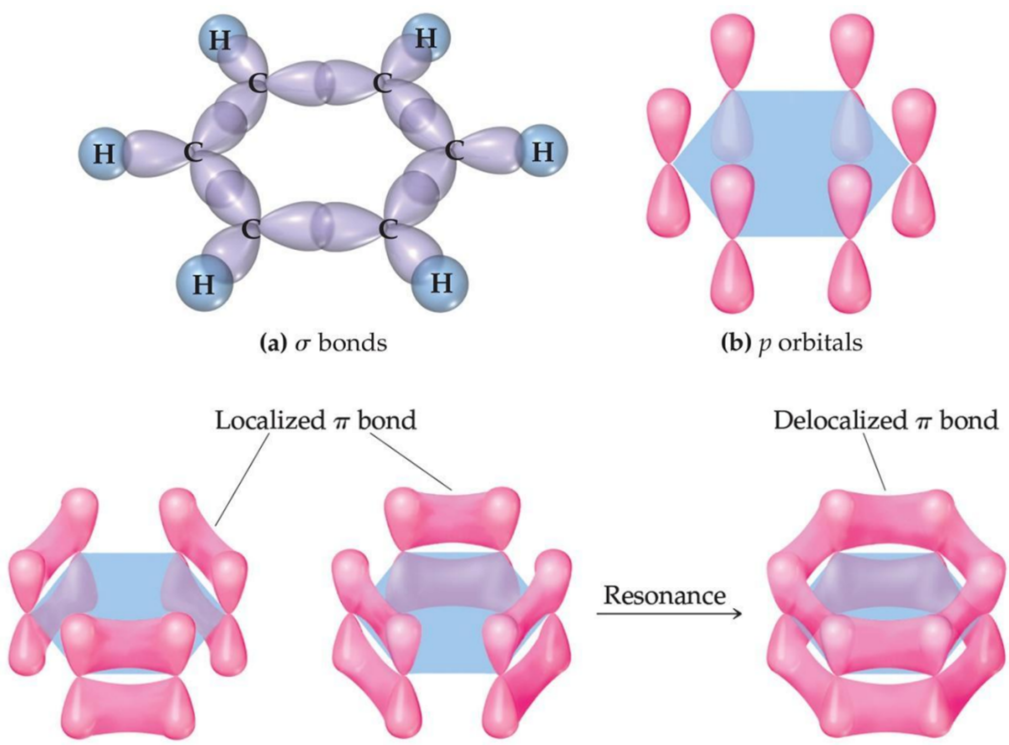
Cis-1,2- dichloroethene

反

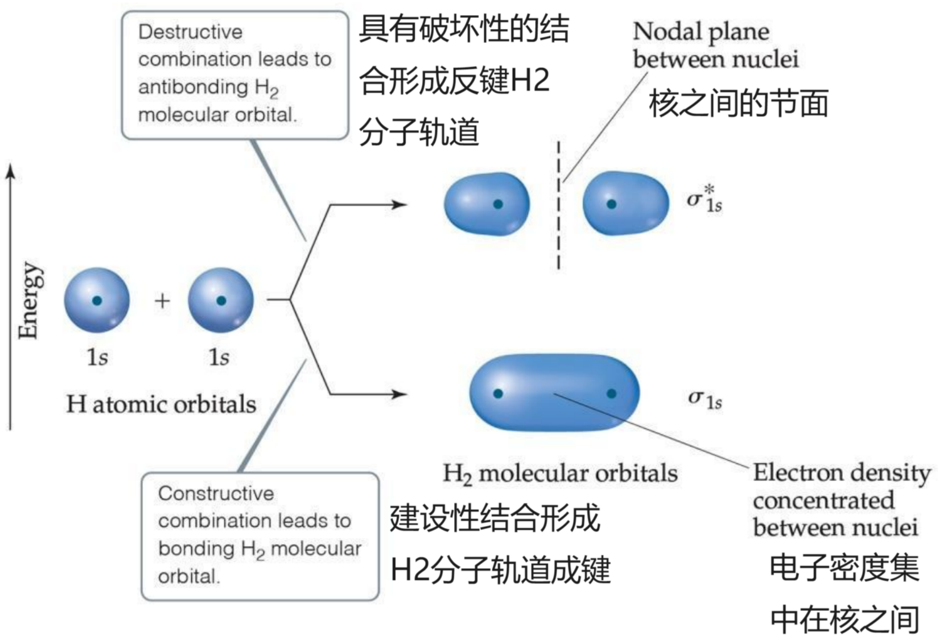


Trans-1,2- dichloroethene

- Localized or Delocalized Electrons定域电子或离域电子
  - Bonding electrons (σ or π) that are specifically shared between two atoms are called localized electrons.两个原子之间共享的键合电子(σ或π)称为定域电子
  - In many molecules, we can't describe all electrons that way (resonance); the other electrons (shared by multiple atoms) are called delocalized electrons.在许多分子中，我们不能这样描述所有的电子(共振);其他电子(由多个原子共享)称为离域电子
  - Benzene苯
    - The organic molecule benzene (C6H6) has six σ bonds and a p orbital on each C atom, which form delocalized bonds using one electron from each p orbital.有机分子苯(C6H6)在每个C原子上有6个σ键和1个p轨道，它们利用每个p轨道的一个电子形成离域键

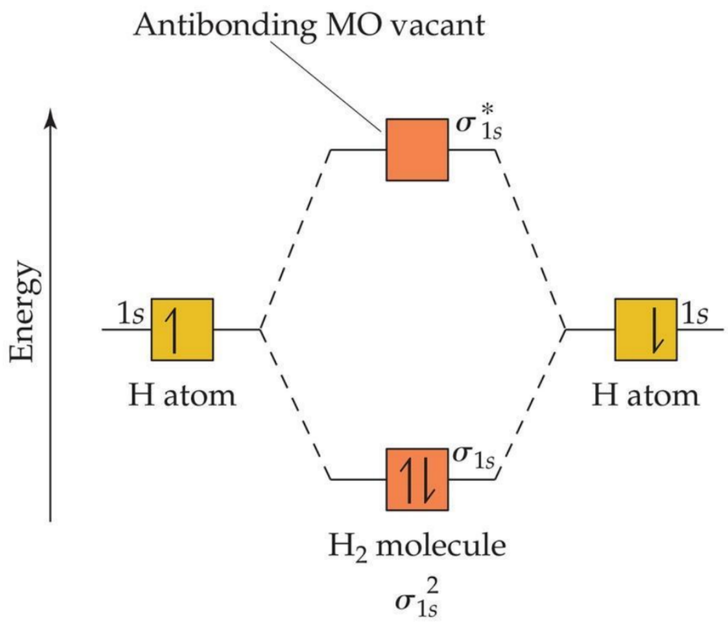


- Molecular Orbital (MO) Theory分子轨道理论
  - Molecular orbitals have many characteristics like atomic orbitals分子轨道和原子轨道一样有很多特点
    - Maximum of two electrons per orbital每个轨道最多两个电子
    - Electrons in the same orbital have opposite spin同一轨道上的电子自旋方向相反
    - Definite energy of orbital轨道的确定能量
    - Can visualize electron density by a contour diagram能用等高线图把电子密度形象化
  - bonding & antibonding
    - Whenever two atomic orbitals overlap, two molecular orbitals are formed: one bonding, one antibonding.每当两个原子轨道重叠时，就会形成两个分子轨道:一个成键，一个反键
    - **Bonding orbitals** are constructive combinations .成键轨道是建设性的组合
    - **Antibonding orbitals** are destructive combinations of atomic orbitals. They have a new feature unseen before: A **nodal plane** occurs where electron density equals zero.反键轨道是原子轨道的破坏性组合，它们有一个以前从未见过的新特征:在电子密度等于零的地方出现了一个节点面
  - σ分子轨道
    - Whenever there is direct overlap of orbitals, forming a bonding and an antibonding orbital, they are called sigma (σ) molecular orbitals. 只要轨道直接重叠，形成成键轨道和反键轨道，它们就被称为sigma (σ)分子轨道
    - The antibonding orbital is distinguished with an asterisk as σ*. 反键轨道用星号表示为σ*
  - Here is an example for the formation of a hydrogen molecule from two atoms.这是一个由两个原子形成氢分子的例子



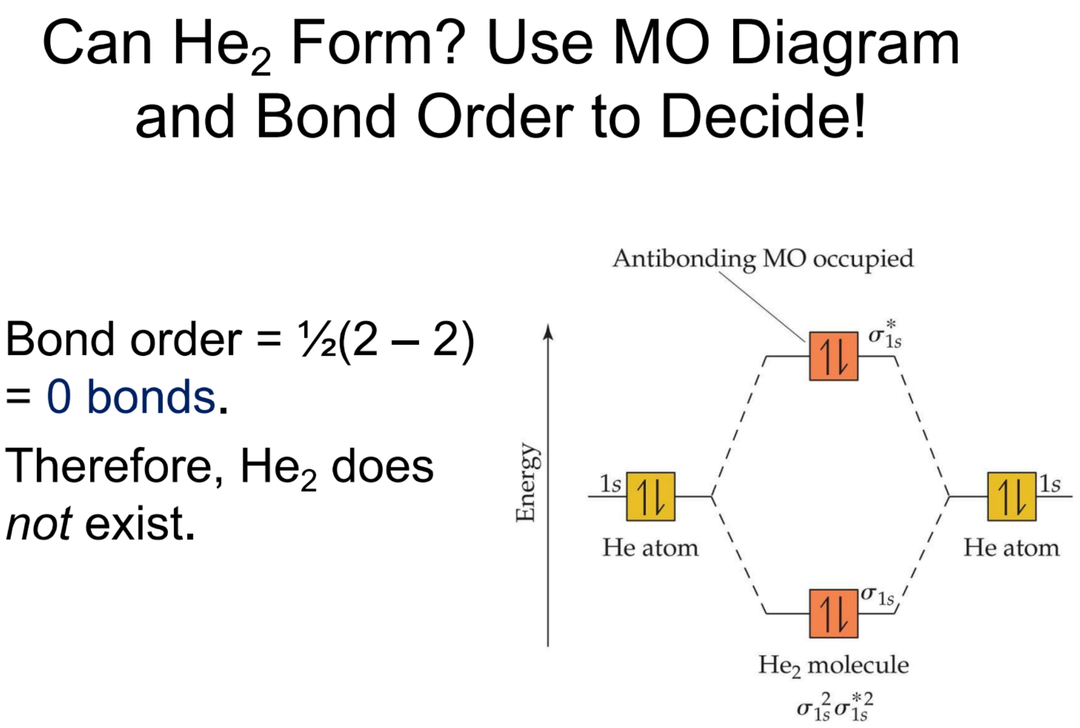
  - MO diagram

- An energy-level diagram, or MO diagram, shows how orbitals from atoms combine能级图，或MO图，显示了原子轨道是如何结合的
- In H2 the two electrons go into the bonding molecular orbital (lower in energy).在H2中，两个电子进入成键的分子轨道(能量较低)



Bond order=1/2(2-0)=1

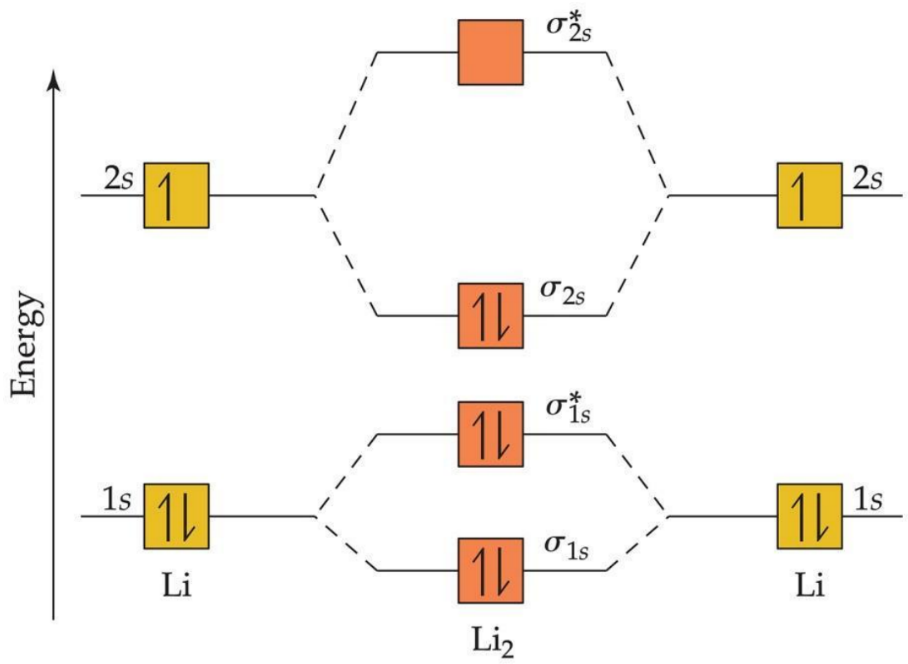
- Bond order = 1/2(number of bonding electrons - number of antibonding electrons)键序=1/2(个成键电子数-反键电子数)
  - 键序愈大，键愈稳定，若键序为零，则不能成键
  - 所以He不能形成分子



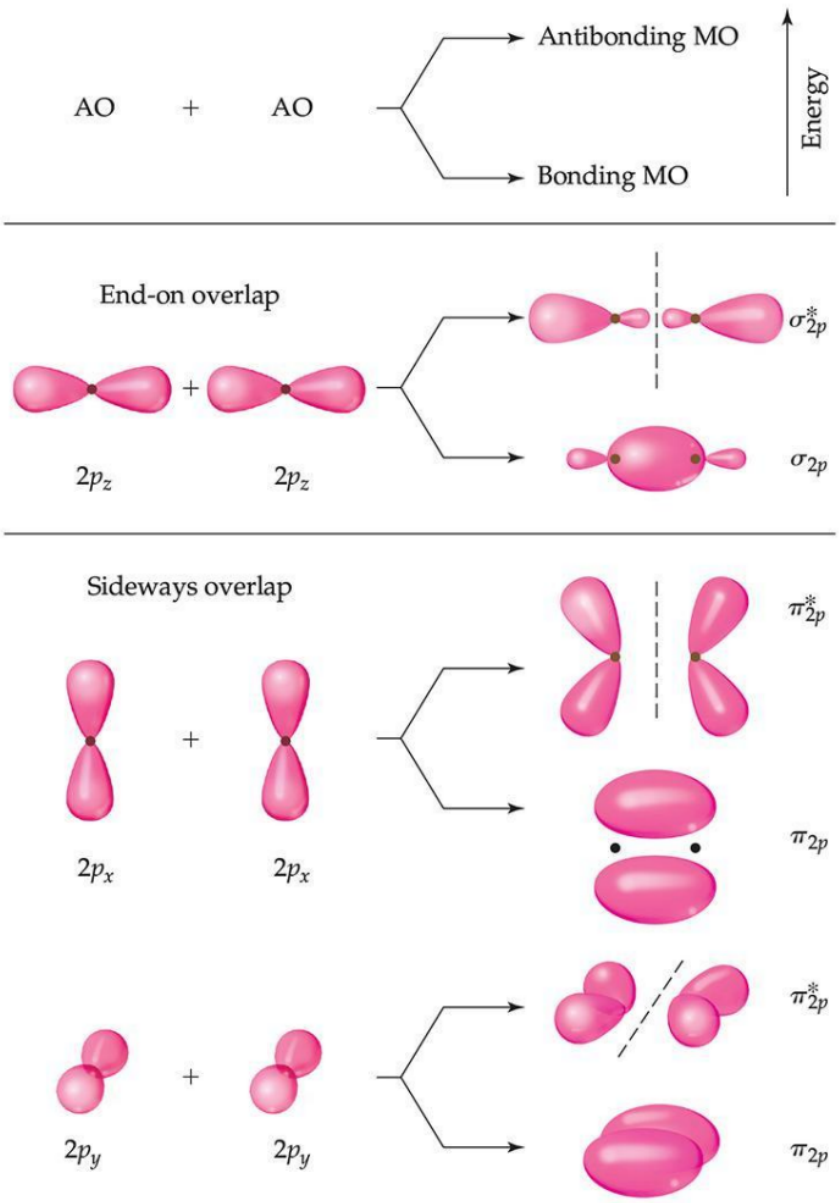
- Guiding Principles for the Formation of Molecular Orbitals分子轨道形成的原则

- The number of MOs formed equals the number of AOs combined.形成的分子轨道的数量等于结合的原子轨道的数量
- AOs combine with AOs of similar energy.原子轨道与能量相近的原子轨道结合
- Each MO can accommodate at most two electrons with opposite spin. (They follow the Pauli exclusion principle.)每个分子轨道最多可以容纳两个自旋相反的电子(它们遵循泡利不相容原理)
- When MOs of the same energy are populated, one electron enters each orbital (same spin) before pairing. (They follow Hund's rules.)当相同能量的MOs被填充时，相同自旋的电子进入不同的轨道(他们遵循洪特规则)

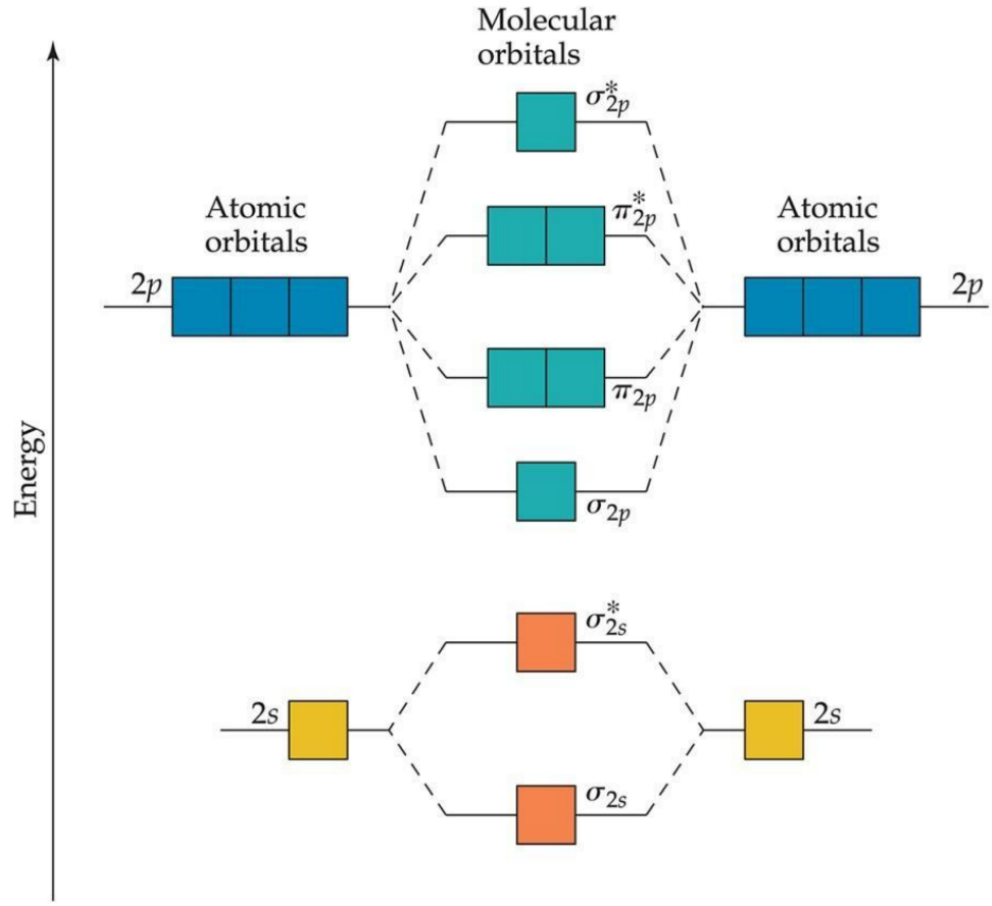
- MOs, Bonding, and Core Electrons分子轨道，成键和核心电子
  - $Li_2(g)$ occurs at high temperatures. $Li_2(g)$在高温下产生
  - Lewis structure: Li—Li 路易斯结构式：Li—Li
  - Notice that core electrons don't play a major part in bonding, so we usually don't include them in the MO diagram.注意内层电子在成键中并不起主要作用，所以我们通常不把它们包括在分子轨道图



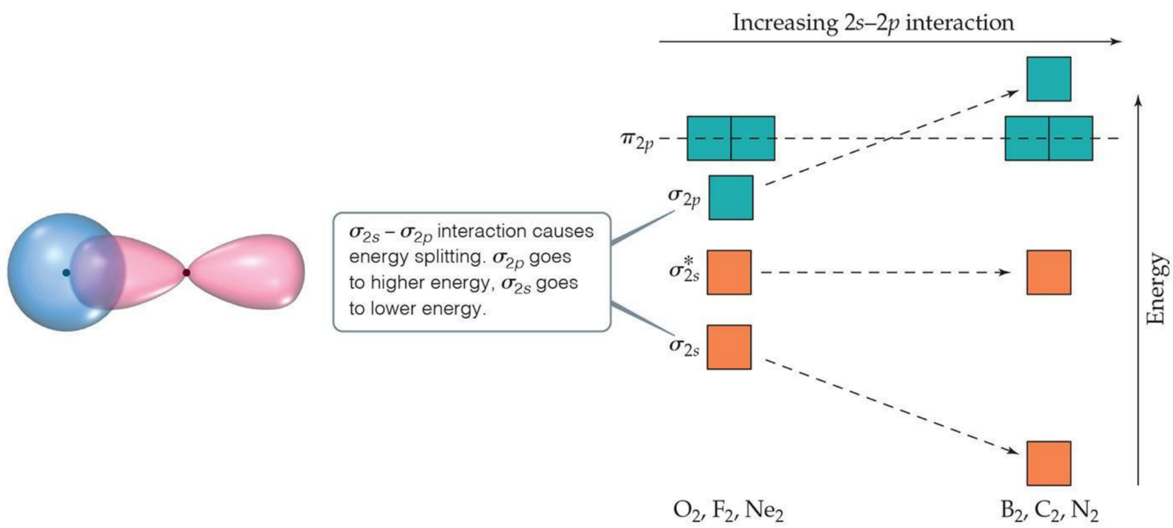
- MOs from p-Orbitals p轨道形成的分子轨道
  - p-orbitals also undergo overlap. p轨道也会发生重叠
  - They result in either direct or sideways overlap.它们导致直接重叠或侧边重叠
  - 图



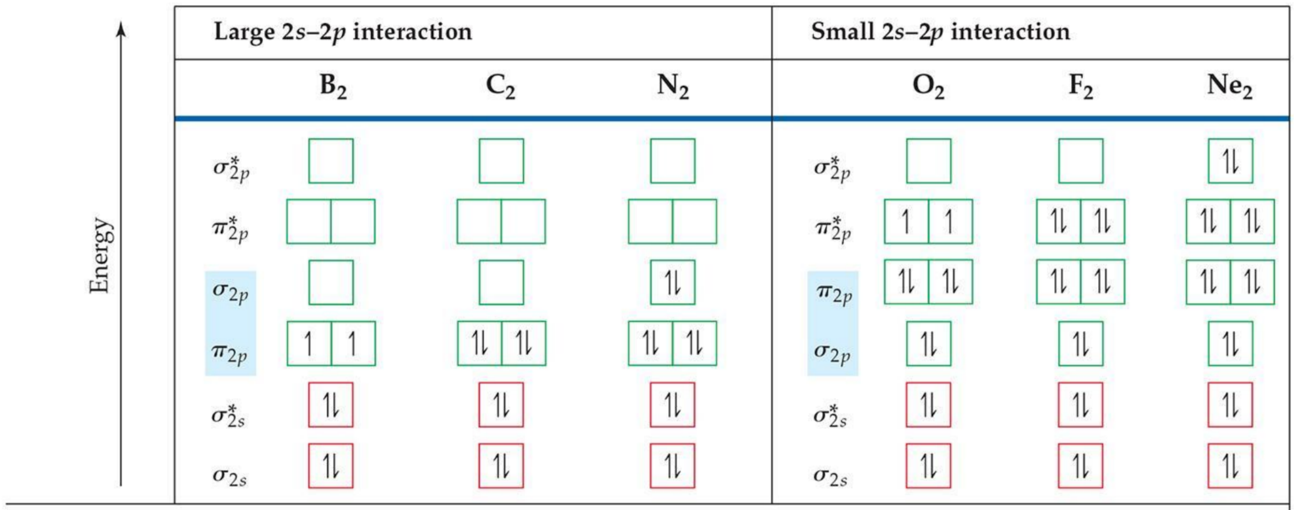
- MO Diagrams for the Second Period p-Block Elements 二周期p区域元素的分子轨道图
  - There are σ and σ* orbitals from s and p atomic orbitals. s和p有σ和σ*轨道
  - There are π and π* orbitals from p atomic orbitals. p原子轨道有π轨道和π*轨道
  - Since direct overlap is stronger, the effect of raising and lowering energy is greater for σ and σ*.由于直接重叠较强，σ和σ*的能量升降效应更大
  - 图



- s and p Orbital Interactions s和p轨道相互作用
    - In some cases, s orbitals can interact with the pz orbitals more than the py and py orbitals.在某些情况下，s轨道比py和py轨道更能与pz轨道相互作用
    - It raises the energy of the pz orbital and lowers the energy of the s orbital.它提高了pz轨道的能量，降低了s轨道的能量
    - The py and py orbitals are degenerate orbitals.py和py轨道是简并轨道
    - 图

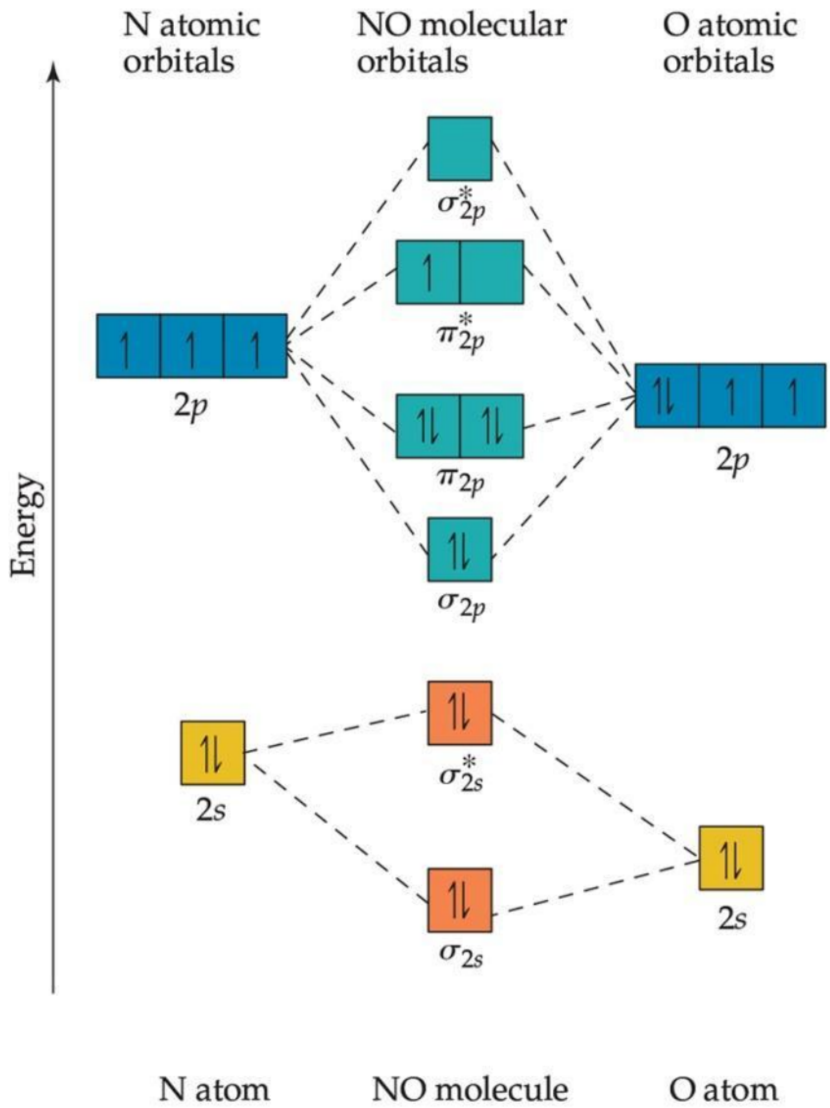


- MO Diagrams for Diatomic Molecules of Second Period Elements第二周期元素双原子分子的MO图



	B_2	C_2	N_2	O_2	F_2	Ne_2
Bond order	1	2	3	2	1	0
Bond enthalpy (kJ/mol)	290	620	941	495	155	—
Bond length (Å)	1.59	1.31	1.10	1.21	1.43	—
Magnetic behavior	Paramagnetic	Diamagnetic	Diamagnetic	Paramagnetic	Diamagnetic	—

- MO Diagrams and Magnetism分子轨道图与磁性
  - **Diamagnetism** is the result of all electrons in every orbital being spin-paired. These substances are weakly repelled by a magnetic field.抗磁性是由于每个轨道上的所有电子都是自旋成对的，这些物质受到磁场的弱排斥，它们产生的磁场方向与外加磁场方向相反，它们的存在会减小外加磁场
  - **Paramagnetism** is the result of the presence of one or more unpaired electrons in an orbital.顺磁性是轨道中存在一个或多个不成对电子的结果，对磁场响应很弱的磁性
  - Paramagnetism of Oxygen氧的顺磁性
    - Lewis structures would not predict that O2 is paramagnetic.路易斯结构不能预测氧是顺磁性的
    - The MO diagram clearly shows that O2 is paramagnetic.MO图清楚地表明O2是顺磁性的
    - Both show a double bond (bond order =2).两者都呈双键(键序=2)
- Heteronuclear Diatomic Molecules异核双原子分子
  - Diatomic molecules can consist of atoms from different elements.双原子分子可以由不同元素的原子组成
  - The atomic orbitals have different energy, so the interactions change slightly.原子轨道有不同的能量，所以相互作用的变化很小
  - The more electronegative atom has orbitals lower in energy, so the bonding orbitals will more resemble them in energy.电负性越强的原子的轨道能量越低，所以成键轨道的能量也就越接近它们
  - 图



N atomic orbitals
NO molecular orbitals
O atomic orbitals
Energy
σ^*_{2p}
π^*_{2p}
2p
π_{2p}
2p
σ_{2p}
σ^*_{2s}
2s
2s
σ_{2s}
N atom
NO molecule
O atom